

VR Education Contents for Core Fundamental Nursing Skills

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ABSTRACT

This educational content concerns fundamental nursing skills by using virtual reality. Blood sugar testing and the administration of insulin subcutaneous injection are two of the most frequently used among the 20 core fundamental nursing skills. The content pertaining to these tasks can be used for self-management education for diabetic patients. This contents designed a core fundamental nursing skill on immersive VR contents by dividing the learner's experience into three stages: guide, mission, and feedback with these skills. And it is designed by tracking the movement of the hand through finger joint recognition without using a controller for immerse in training. This content will help improve nursing students' clinical practice competency.

CCS CONCEPTS

• Human-centered computing-> VR;

KEYWORDS

Virtual Reality, Core Fundamental Nursing Skills, VR Education Contents, Finger Joint Recognition

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1 INTRODUCTION

One of the fundamental nursing practices in the curriculum entails learning the theory and procedures related to the 20 core fundamental nursing skills in order to solve various patients' nursing problems. Fundamental nursing skills VR content was created to help learners gain self-directed problem-solving experiences using PBL(Problem Based Learning), the most basic and widely used nursing educational method via in VR. We selected blood sugar testing and insulin subcutaneous injection as topics for which to attempt to create realistic and immersive VR nursing education content because they are frequently used and important. This content is called core fundamental nursing skills in immersive VR, which is abbreviated to CFNS-iVR.

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Figure 1: Screen of CFNS-iVR

2 OUR APPROACH

We made this training in the form of a gaming mission, so skills are trained as students complete 24 missions successfully and accurately on time. Our system differs from Smart Nursing of gridatech in that students are scored for each mission. To make the VR space more realistic, three interior spaces (nursing station, patient room, and treatment room) were created based on photos of a laboratory. To improve students' communication skills we created a character who is a nurse in their 20s and another who is a patient with diabetes in their mid-50s (within the age range of most diabetic patients who require blood sugar measurement and insulin subcutaneous injection). Additionally, to communicate with patients, we combined our content with the Google's speech-to-text (STT) voice recognition function. Moreover, most of the existing VR education content relies heavily on a controller like an Asan Hostital VR education system. There exists a huge discrepancy between learning nursing skills using a controller and treating patients in real life. To overcome this limitation, our content use real hand movements instead of using a controller in Oculus Quest 2 (Oculus Corp, 2021). And we use the Same Corp's Mollisen HAND to live capture finger joint movements and output animation data for following precise movements. Our system was actually used in the School of nursing at Soonchunhyang University in 2021.

3 TRAINING SCENARIO DESIGN

The composition of CFNS-iVR is as follows: guidance as the first step, simulation practice through missions as the second, and feedback as the third step. The first step is to guide learners to perceive the VR as if they can subjectively move around. The second step, in simulation practice, a pop-up animation is used to guide the correct movements for a particular mission or patient character to accomplish the mission successfully. There are four different



Figure 2: Demonstration of CFNS-iVR

locations, as if they are working in real space in real life. The third step is called feedback. In this step, they submit their results and reflect on their performance.

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